

2016.0 RANGE ROVER (LG), 303-03

ENGINE COOLING – TDV6

3.0L DIESEL – GEN 2/TDV6

3.0L DIESEL

DIAGNOSIS AND TESTING

PRINCIPLE OF OPERATION

For a detailed description of the engine cooling system and operation, refer to the relevant Description and Operation section of the workshop manual.

REFER to: [Engine Cooling](#) (303-03A Engine Cooling - TDV6 3.0L Diesel - Gen 2/TDV6 3.0L Diesel, Description and Operation).

INSPECTION AND VERIFICATION

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault and may also cause additional faults in the vehicle being checked and/or the donor vehicle.

NOTE:

Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.

1. Verify the customer concern.
2. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Coolant leaks ▪ Coolant hoses ▪ Coolant expansion tank ▪ Radiator ▪ Heater core ▪ Active grill air shutter ▪ Accessory drive belt ▪ Cooling fan	▪ Fuses ▪ Harnesses ▪ Loose or corroded connector(s) ▪ Engine coolant temperature sensor

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.
4. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index.

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Coolant loss	▪ Hoses ▪ Hose connections	Carry out a visual inspection. If there are no obvious leaks, carry out a cooling system pressure

	▪ Radiator ▪ Water pump ▪ Heater core ▪ Gaskets ▪ Engine casting cracks ▪ Engine block core plugs	test. Rectify any leaks as necessary. Refer to the relevant section of the workshop manual
Overheating	▪ Low/contaminated coolant ▪ Thermostat ▪ Cooling fan ▪ Engine coolant temperature sensor ▪ Restricted air flow over the radiator ▪ Active grill air shutter	Check the coolant level and condition. Carry out a cooling system pressure test. Rectify any leaks as necessary. Check the thermostat and rectify as necessary. Check the cooling fan operation, make sure the cooling fan rotates freely. Check for obstructions to the air flow over the radiator. Check the active grill air shutter operation using the manufacturer approved service tool
Engine not reaching normal temperature	▪ Thermostat ▪ Cooling fan	Check the thermostat operation. Check the cooling fan operation, make sure the cooling fan is not seized

DTC INDEX

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00.

REFER to: [Diagnostic Trouble Code Index - TDV6 3.0L Diesel - Gen 1.5, DTC: Engine Control Module \(ECM\)](#) (100-00 General Information, Description and Operation).